Explanation of Quantum Chaotic Dynamics in Matrix Gauge Theories

PhD Thesis Committee Report

Berk Özcan
Supervisor: Prof. Dr. Seçkin Kürkçüoğlu

Middle East Technical University
Department of Physics

May 2024 - May 2027

Outline

1. Introduction & Motivation
2. Model and Methods
3. Computational Methods & Results
4. Conclusion

1. Introduction & Motivation
2. Model and Methods
3. Computational Methods & Results
4. Conclusion

Motivation

- **Gauge-Gravity Duality:** Connections established between chaotic systems and black hole physics have motivated research into chaotic dynamics in quantum systems.
- **Measuring Chaos:**
  - Classical: Lyapunov exponents (exponential divergence of trajectories).
  - Quantum: Out-of-Time-Ordered Correlators (OTOCs) where $C(t) \sim e^{2\lambda_L t}$.
- **The MSS Bound:** Maldacena, Shenker, and Stanford suggested a universal upper bound on quantum chaos:

$$\lambda_L \leq \frac{2\pi k_B T}{\hbar}$$

Black holes are expected to saturate this bound.

Thesis Goals

Primary Goal

To study the quantum level dynamics, starting from near thermal equilibrium configurations in Bosonic Matrix (BM) models on Fuzzy 2-Spheres ($S_F^2 \times \mathbb{R}^1$).

Specific Objectives

1. Apply **Gaussian State Approximation (GSA)** to average Heisenberg equations of motion to obtain the equations of motion for the 1-point and 2-point correlators.
2. Solve the resulting truncated differential equations numerically.
3. Calculate the **Largest Lyapunov Exponent (λ_L)** to determine chaotic behavior.
4. Investigate the relationship between λ_L, Energy, and Temperature to check the MSS bound.

1 Introduction & Motivation

2 Model and Methods

3 Computational Methods & Results

4 Conclusion

Hamiltonian on $S_F^2 \times \mathbb{R}$

The Hamiltonian for the matrix models considered is expressed as:

$$\hat{H} = \frac{1}{2} Tr \left[\hat{P}_i^2 + \mu^2 \hat{X}_i^2 - \frac{[L_a, \hat{X}_i]^2}{R^2} - \frac{\lambda}{2} \sum_{i \neq j} [\hat{X}_i, \hat{X}_j]^2 \right]$$

- $\hat{X}_i$: $N \times N$ Hermitian matrices.
- $\hat{P}_i$: $N \times N$ Hermitian conjugate momentum operators i.e.

$$[\hat{X}_i, \hat{P}_j] = i\delta_{ij} \mathbb{I}_{N^2 \times N^2}.$$

- $\mathcal{L}_a := [L_a, \cdot]$: Derivatives on S_F^2 satisfying $SU(2)$ algebra.
- L_a: Spin $\frac{N-1}{2}$ IRR of $SU(2)$.

The system exhibits $SU(2)$ symmetry and, when planar/non-planar terms are included, $SO(2)$ symmetry and D_4 when non-planar terms are excluded.

Real Polarization Operator Basis

To facilitate numerical solution, the Hamiltonian is expanded using the real polarization operator basis Y_{lm}:

$$\hat{X}_i = \hat{X}_i^{lm} Y_{lm}, \quad \hat{P}_i = \hat{P}_i^{lm} Y_{lm}.$$

Which is given in terms of the Polarization Operator Basis T_{lm} as:

$$Y_{lm} = \begin{cases} \frac{1}{\sqrt{2}}(T_{lm} + T_{lm}^\dagger) & \text{if } m > 0 \\ T_{l0} & \text{if } m = 0 \\ \frac{i}{\sqrt{2}}(T_{lm} - T_{lm}^\dagger) & \text{if } m < 0 \end{cases},$$

with the derivatives on S_F^2 given by:

$$\mathcal{L}_3 Y_{lm} = [L_3, Y_{lm}] = m Y_{lm}$$

$$\mathcal{L}^2 Y_{lm} = [L_a, [L_a, Y_{lm}]] = -[L_a, Y_{lm}]^2 = l(l+1) Y_{lm}$$

Hamiltonian in Real Polarization Operator Basis

Using the real polarization operator basis in the Hamiltonian allows us to write it in terms of trace coefficients ($\mathcal{K}$ symbols):

$$H = \frac{1}{2}\Bigg[\left(P_i^{lm}\right)^2 + \sum_{l=0}^{N-1}\left(\mu^2 + \frac{l(l+1)}{R^2}\right)\left(X_i^{lm}\right)^2$$

$$+ \lambda \mathcal{K}^{l_1 l_2 l_3 l_4}_{m_1 m_2 m_3 m_4}\left(X_1^{l_1 m_1} X_1^{l_2 m_2} X_2^{l_3 m_3} X_2^{l_4 m_4} - X_1^{l_1 m_1} X_2^{l_2 m_2} X_1^{l_3 m_3} X_2^{l_4 m_4}\right)\Bigg]$$

$$Tr\left(Y_{l_1 m_1} Y_{l_2 m_2} Y_{l_3 m_3} Y_{l_4 m_4}\right)$$

Heisenberg Equations of Motion

We can compute the Heisenberg equations of motion using

$$\dot{\xi}_i^{l_1 m_1} = i[H, \xi_i^{l_1 m_1}], \text{ with } [X_i^{l_1 m_1}, P_j^{l_2 m_2}] = i\delta_{ij}\delta^{l_1 l_2}\delta^{m_1 m_2}$$

This gives us the following equations of motion:

$$\dot{X}_1^{lm} = P_1^{lm}, \; \dot{X}_2^{lm} = P_2^{lm},$$

$$\dot{P}_1^{lm} = -\left[\left(\mu^2 + \frac{l(l+1)}{R^2}\right) X_1^{lm} + \frac{\lambda}{2}\mathcal{N}^{l\; l_1 l_2 l_3}_{m\; m_1 m_2 m_3} X_1^{l_1 m_1} X_2^{l_2 m_2} X_2^{l_3 m_3}\right]$$

$$\dot{P}_2^{lm} = -\left[\left(\mu^2 + \frac{l(l+1)}{R^2}\right) X_1^{lm} + \frac{\lambda}{2}\underbrace{\mathcal{N}^{l\; l_1 l_2 l_3}_{m\; m_1 m_2 m_3}}_{\substack{\mathrm{Tr}\left(Y_{lm}Y_{l_1 m_1}Y_{l_2 m_2}Y_{l_3 m_3}\right) + \mathrm{Tr}\left(Y_{l_1 m_1}Y_{lm}Y_{l_2 m_2}Y_{l_3 m_3}\right) \\ -\mathrm{Tr}\left(Y_{lm}Y_{l_2 m_2}Y_{l_1 m_1}Y_{l_3 m_3}\right) - \mathrm{Tr}\left(Y_{l_1 m_1}Y_{l_2 m_2}Y_{lm}Y_{l_3 m_3}\right)}} X_2^{l_1 m_1} X_1^{l_2 m_2} X_1^{l_3 m_3}\right].$$

Gaussian State Approximation (GSA)

To inspect the real time dynamics of the system, we use the Gaussian State Approximation (GSA) to average the Heisenberg equations of motion, using the following Wigner function of the Gaussian density matrix:

$$W(\xi) = \mathcal{N} e^{-\frac{1}{2}(\xi - \langle\xi\rangle)^T \Sigma^{-1} (\xi - \langle\xi\rangle)}$$

where the covariance matrix is given by:

$$\Sigma = \begin{pmatrix} \langle\langle X_i^{lm} X_j^{l'm'} \rangle\rangle & \langle\langle X_i^{lm} P_j^{l'm'} \rangle\rangle \\ \langle\langle X_i^{lm} P_j^{l'm'} \rangle\rangle & \langle\langle P_i^{lm} P_j^{l'm'} \rangle\rangle \end{pmatrix}$$

- **Wick's Theorem:** Used to express the equations of motion via 1-point and 2-point correlation functions.
- **Result:** A coupled system of truncated differential equations for $\langle X \rangle$, $\langle P \rangle$, and correlators like $\langle XX \rangle$, $\langle XP \rangle$ and $\langle PP \rangle$.

Gaussian Averaged EOM

The equations of motion for the correlators are derived via Gaussian averaging. For example:

$$\partial_t \langle X_1^{lm} \rangle = \langle P_1^{lm} \rangle$$
$$\partial_t \langle P_1^{lm} \rangle = -\left[\left(\mu^2 + \frac{l(l+1)}{R^2} \right) \langle X_1^{lm} \rangle + \frac{\lambda}{2} \mathcal{N} \ldots \right]$$

The dynamics of the 2-point correlators (e.g., $\partial_t \langle\langle XX \rangle\rangle$) are similarly derived, recovering classical equations if 2-point correlators are zero.

Thermal States

To construct the thermal states, we need to use the symmetries of the system to find the appropriate initial conditions.

- **Symmetry:** Parity and Time Reversal symmetry.
- **Result:**

$$\langle X_1^{lm} \rangle = \langle X_2^{lm} \rangle = 0,\ \langle P_1^{lm} \rangle = \langle P_2^{lm} \rangle = 0,\ \langle X_i^{lm} P_j^{l'm'} \rangle = 0$$

- **Symmetry:** $SO(2)$ or D_4 symmetry.
- **Result:**

$$\langle X_i^{lm} X_j^{lm} \rangle \sim \delta_{ij}$$

- **Symmetry:** $SU(2)$ symmetry.
- **Result:**

$$\langle X_i^{lm} X_j^{l'm'} \rangle = \sigma_{xx}(l)\delta_{ij}\delta^{ll'}\delta^{mm'},\ \langle P_i^{lm} P_j^{l'm'} \rangle = \sigma_{pp}(l)\delta_{ij}\delta^{ll'}\delta^{mm'}$$

Entropy

- von Neumann entropy for Gaussian density matrix can be written in terms of the symplectic eigenvalues.
- Symplectic eigenvalues can be written using the eigenvalues of the $\Sigma\Omega$ matrix. (Ω: Symplectic form of canonical coordinates.)
- Eigenvalues of the matrix come in complex pairs $\lambda_{[k]} = \pm i f_{[k]}$, where the $f_{[k]}$ are the symplectic eigenvalues.

$$S = \sum_{[k]} \left(f_{[k]} + \frac{1}{2} \right) \ln \left(f_{[k]} + \frac{1}{2} \right) - \sum_{[k]} \left(f_{[k]} - \frac{1}{2} \right) \ln \left(f_{[k]} - \frac{1}{2} \right),$$

$$= \sum_{l} (2l+1) \left[\left(f_l + \frac{1}{2} \right) \ln \left(f_l + \frac{1}{2} \right) - \left(f_l - \frac{1}{2} \right) \ln \left(f_l - \frac{1}{2} \right) \right]$$

Symplectic Eigenvalues

- Using these initial conditions we can write the entropy from the symplectic matrix

$$
\begin{aligned}
(\Sigma\Omega)_{ij}^{[lm][l'm']} &= \Sigma_{ik}^{[lm][l''m'']}\Omega_{kj}^{[l''m'']}{}^{[l'm']} \\
&= \begin{pmatrix} \sigma_{xx}(l)\delta^{ll'}\delta^{mm'}\delta_{ik} & 0 \\ 0 & \sigma_{pp}(l)\delta^{ll'}\delta^{mm'}\delta_{ik} \end{pmatrix} \begin{pmatrix} 0 & \delta^{l''l'}\delta^{m''m'}\delta_{kj} \\ -\delta^{l''l'}\delta^{m''m'}\delta_{kj} & 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 & \sigma_{xx}(l)\delta^{ll'}\delta^{mm'}\delta_{ij} \\ -\sigma_{pp}(l)\delta^{ll'}\delta^{mm'}\delta_{ij} & 0 \end{pmatrix}.
\end{aligned}
$$

- The symplectic eigenvalues are then

$$f_{[k]} = \pm\sqrt{\sigma_{xx}(l)\sigma_{pp}(l)}.$$

- $[k] \equiv [i, l, m] = 1, 2, \ldots, 2N^2$. With only N independent symplectic eigenvalues.

Energy

The energy of the system is given by $E = \langle H \rangle$. If we use the initial conditions given previously, we get:

$$E = \sum_{l=0}^{N-1} (2l+1) \left(\sigma_{pp}(l) + \left(\mu^2 + \frac{l(l+1)}{R^2} \right) \sigma_{xx}(l) \right) + \frac{\lambda C^2}{2N}$$

Here C is

$$C := \sum_{l=0}^{N-1} (2l+1) \sigma_{xx}(l) = \frac{\langle\langle Tr\,(X_i X_i) \rangle\rangle}{2}$$

Entropy Maximization

We need to maximize the Entropy S for a fixed energy E. We can use the method of Lagrange multipliers to find the maximum of

$$\mathcal{L} = S + \beta\left(E - E_0\right)$$

We can maximize the entropy by taking the derivative of $\mathcal{L}$ with respect to $\sigma_{xx}(l)$ and $\sigma_{pp}(l)$ and setting it to zero, we get:

$$\frac{\sigma_{pp}(l)}{\sigma_{xx}(l)} = \frac{\omega_l}{N^2}$$

where $\omega_l := N\sqrt{\mu^2 + \frac{l(l+1)}{R^2} + \frac{\lambda C}{N}}$. We also have the symplectic eigenvalue equation:

$$\sigma_{xx}(l)\sigma_{pp}(l) = f_l^2$$

which can be solved to give

$$f_l = \frac{1}{2} coth\left(\frac{\beta\omega_l}{N}\right)$$

Consistency Equation

Now we know that the symplectic eigenvalues for the maximum entropy state, all we have to do is to find the value of C for a given β and system parameters.

Consistency Equation

The relationship between the C value and temperature β is determined by solving:

$$C = \frac{1}{2}\sum_{l=0}^{N-1}(2l+1)\frac{\coth(\beta\omega_l/N)}{\omega_l}$$

where ω_l depends on μ, R, and C itself.

Solving this equation allows us to exactly determine f_l for a given β and system parameters.

Selecting Initial Conditions

Static Solution

By plugging in the values $\sigma_{xx}(l) = \frac{N f_l}{\omega_l}$ and $\sigma_{pp}(l) = \frac{\omega_l f_l}{N}$ in our equations of motion, we can check the Thermal equilibrium is also the static solution of the system.

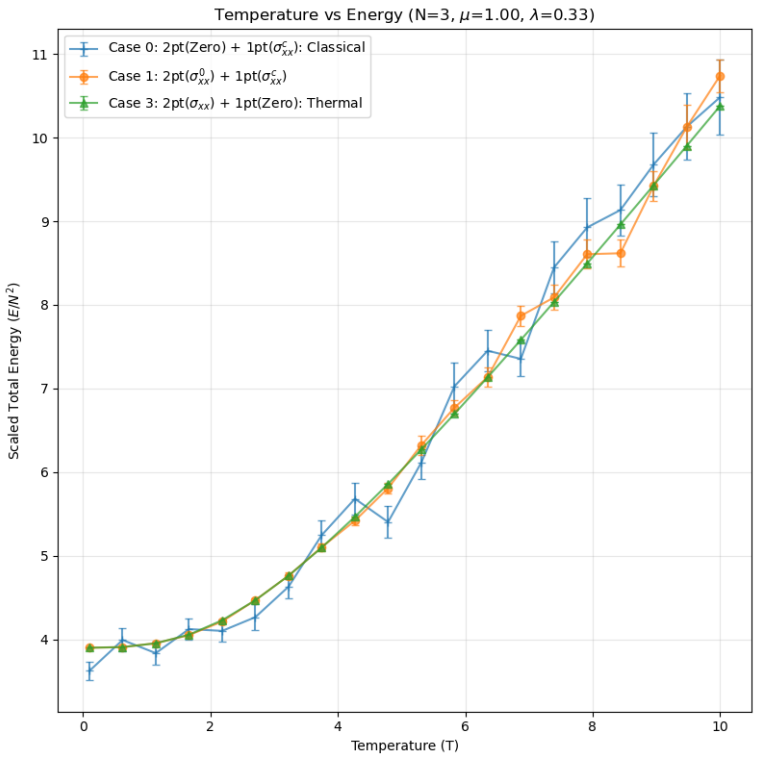
Therefore, we need a way to choose initial conditions while also ensuring the system is dynamical. To do so, we can separate the variance into quantum and classical dispersions:

$$\sigma_{\xi\xi}(l) = \sigma^0_{\xi\xi}(l) + \sigma^c_{\xi\xi}(l)$$

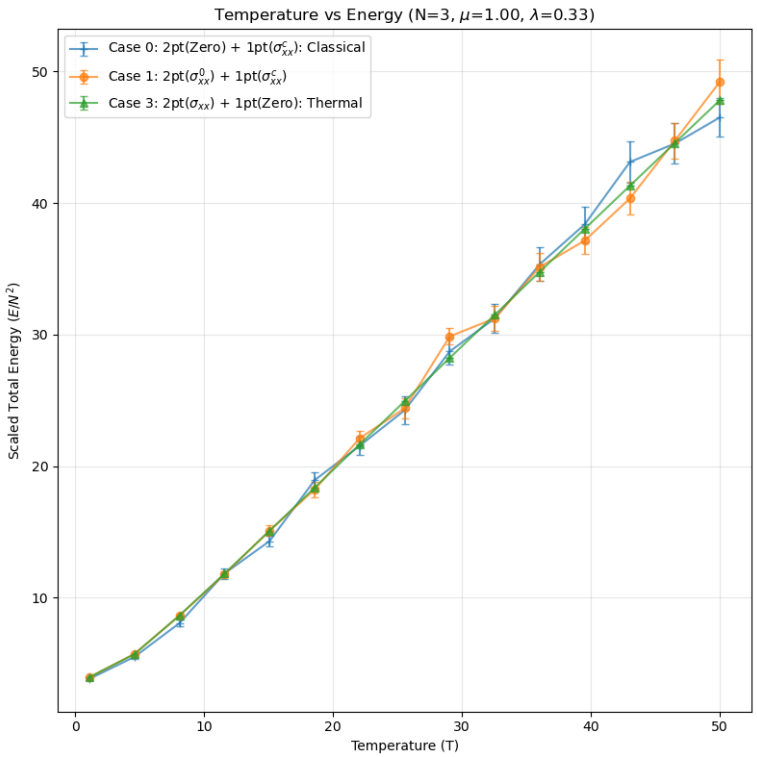
- **Quantum Dispersion (σ^0):** Zero-temperature limit.
- **Classical Dispersion (σ^c):** Difference between thermal and zero-temp limits.

One-point initial conditions are selected randomly from a Gaussian distribution with variance σ^c to ensure the system is dynamical.

Energy vs Temperature



(a) T values from 0.01 to 10



(b) T values from 0.01 to 50

Figure 1: Energy vs Temperature for $N = 3, R = 0.5, \lambda = \frac{1}{N}, \mu = \frac{1}{2R}$.

1 Introduction & Motivation

2 Model and Methods

3 Computational Methods & Results

4 Conclusion

Computational Methods

- **Implementation:** C++ code structure involving `Analysis`, `Systems`, and `LyapunovAnalysis` modules.
- **Integration Scheme:** Symplectic Euler Method.
  - Preserves the symplectic structure of the system.
  - Update order: $P \to XP \to PP \to XX \to X$.
- **Lyapunov Calculation:** Renormalize-and-sum procedure.

$$\lambda_L \approx \frac{1}{n\Delta t} \sum_{i=1}^{n} \ln \left(\frac{|\delta\xi(t_i)|}{|\delta\xi(0)|} \right)$$

Computational Methods: Integration Scheme

The integration scheme is given by the following equations:

$$
\begin{aligned}
\langle P\rangle_{n+1} &= \langle P\rangle_n \quad + hF^{(2)}\left(\langle X\rangle_n, \langle\langle XX\rangle\rangle_n\right) \\
\langle XP\rangle_{n+1} &= \langle XP\rangle_n + hF^{(4)}\left(\langle X\rangle_n, \langle\langle XX\rangle\rangle_n, \langle\langle PP\rangle\rangle_n\right) \\
\langle PP\rangle_{n+1} &= \langle PP\rangle_n + hF^{(5)}\left(\langle X\rangle_n, \langle\langle XX\rangle\rangle_n, \langle\langle XP\rangle\rangle_{n+1}\right) \\
\langle XX\rangle_{n+1} &= \langle XX\rangle_n + hF^{(3)}\left(\langle\langle XP\rangle\rangle_{n+1}\right) \\
\langle X\rangle_{n+1} &= \langle X\rangle_n \quad + hF^{(1)}\left(\langle P\rangle_{n+1}\right)
\end{aligned}
$$

This leads to the better conservation of the symplectic structure of the system compared to the Runge-Kutta method.

Computational Methods: Lyapunov Calculation

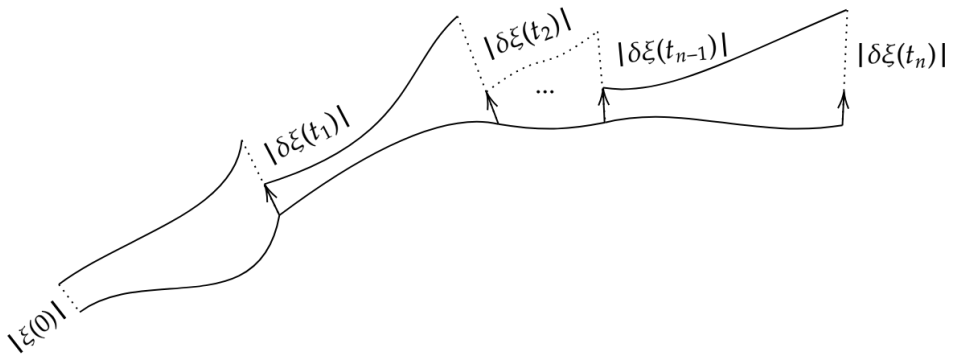


Figure 2: Lyapunov Exponent Computation.

- Integrate two nearby trajectories
- Measure their separation $\delta\xi(t)$
- Renormalize and accumulate growth:

$$\lambda_L = \lim_{t\to\infty} \frac{1}{t} \ln \frac{|\delta\xi(t)|}{|\delta\xi(0)|} \to \lambda_L \approx \frac{1}{n\Delta t} \sum_{i=1}^{n} \ln \frac{|\delta\xi(t_i)|}{|\delta\xi_0|}.$$

λ_L vs Temperature

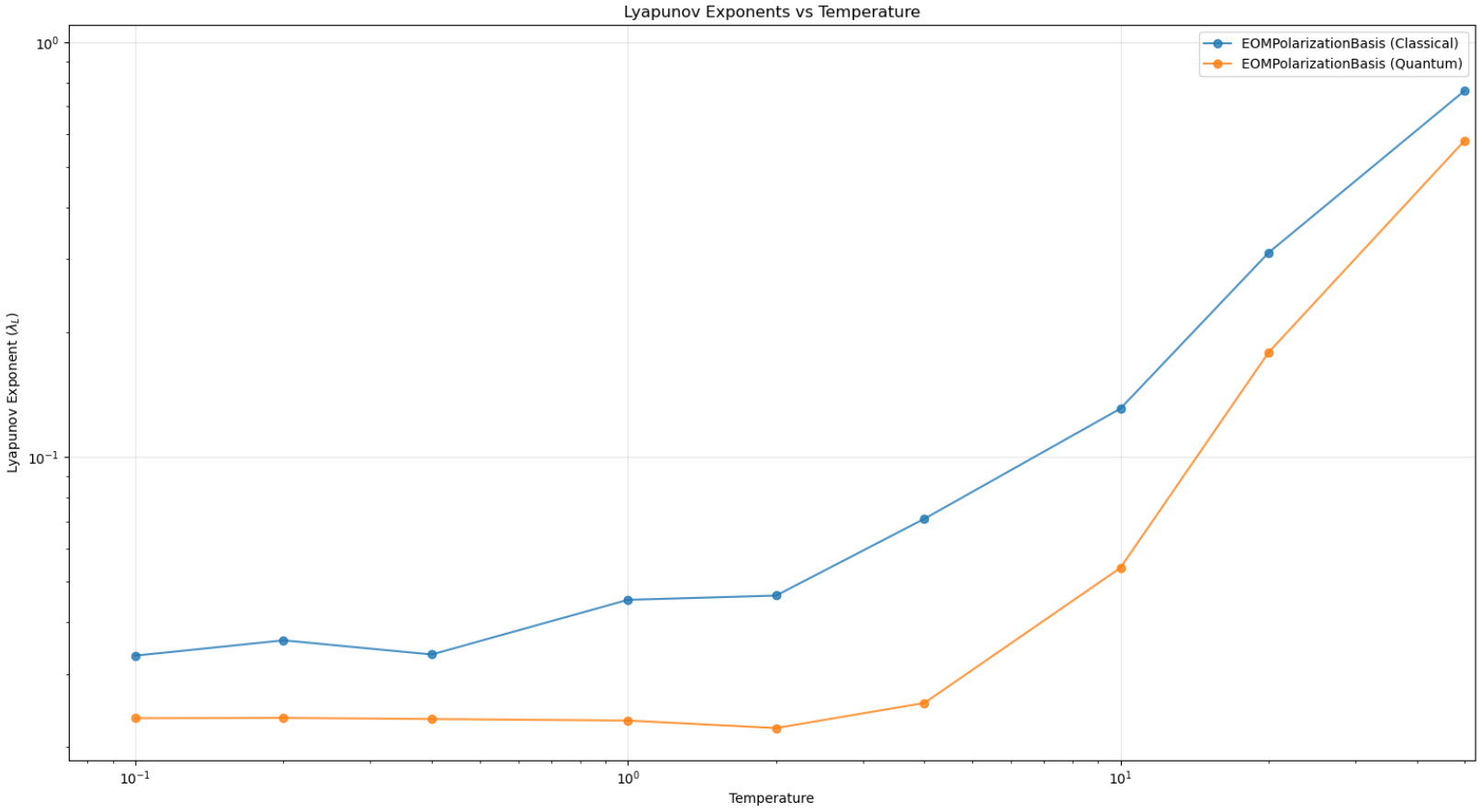


Figure 3: λ_L vs Temperature for $N = 3, R = 0.5, \mu = 1, \lambda = 1/3$.

Results: Lyapunov Exponents

Calculations performed for $N = 3, R = 0.5, \mu = 1, \lambda = 1/3$ at varying temperatures.

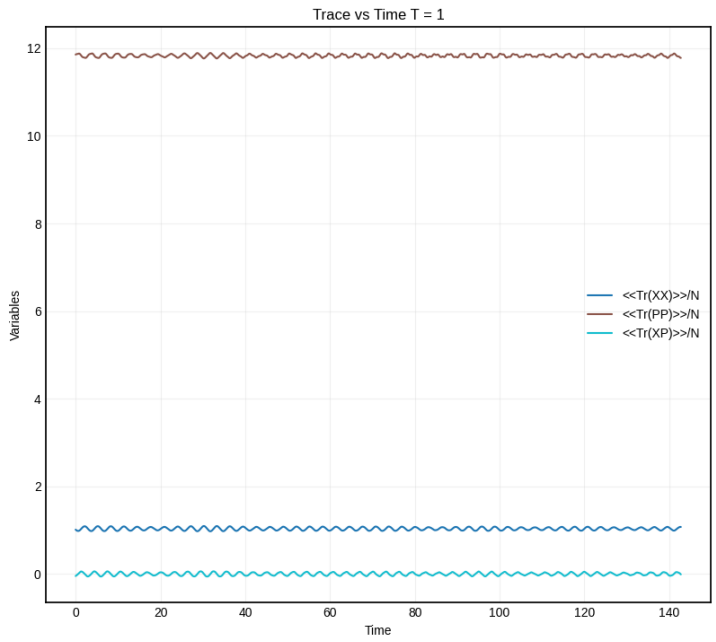
Findings:

- λ_L is positive for $T \geq 0.1$, indicating **Chaos**.
- λ_L increases with temperature.
- Results are consistent with the MSS bound.

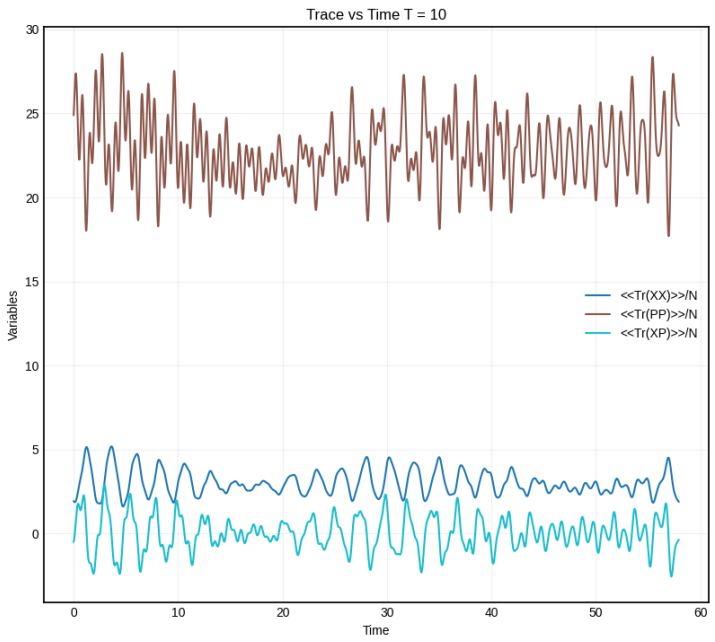
Observation

- Higher temperatures lead to faster divergence and increased stiffness in the system dynamics.
- Increasing the matrix size N lead to a less chaotic system due to t'Hooft scaling.

Results: Time Evolution



(a) $T = 1$



(b) $T = 10$

- **Stability:** The maximum simulation time decreases as temperature increases due to the numerical instability of the system.
- **Interpretation:** The system becomes more chaotic and stiff at high T.

1 Introduction & Motivation

2 Model and Methods

3 Computational Methods & Results

4 Conclusion

Conclusion & Future Work

Summary:

- Developed a framework for real-time quantum dynamics of Matrix Models using Gaussian State Approximation.
- Successfully implemented Symplectic Euler integration for coupled 1-point and 2-point correlators.
- Verified chaotic nature ($\lambda_L > 0$) and thermal consistency.

Future Outlook:

- Further investigate the bosonic part of the BMN model (mass deformation of BFSS).
- Investigate quantum information features (entanglement).